

Emotion-Aware Song Recommendation System

Using ResNet-50 for Mood Detection and K-Means Clustering for Song Classification

Toronto
Metropolitan
University

Termeh Mohebbie
Rehan Ashfaq Syed
Jessie Ma
Harsh Panchal

Deep Learning Course Project - Fall 2024



Why Emotion-Based Music Recommendation?

Significance of Music:

- Universal tool for emotional regulation and expression
- Influences mood, productivity, and stress levels

Challenges in Existing Systems:

- Reliance on historical user data and preferences
- Limited responsiveness to real-time emotional states

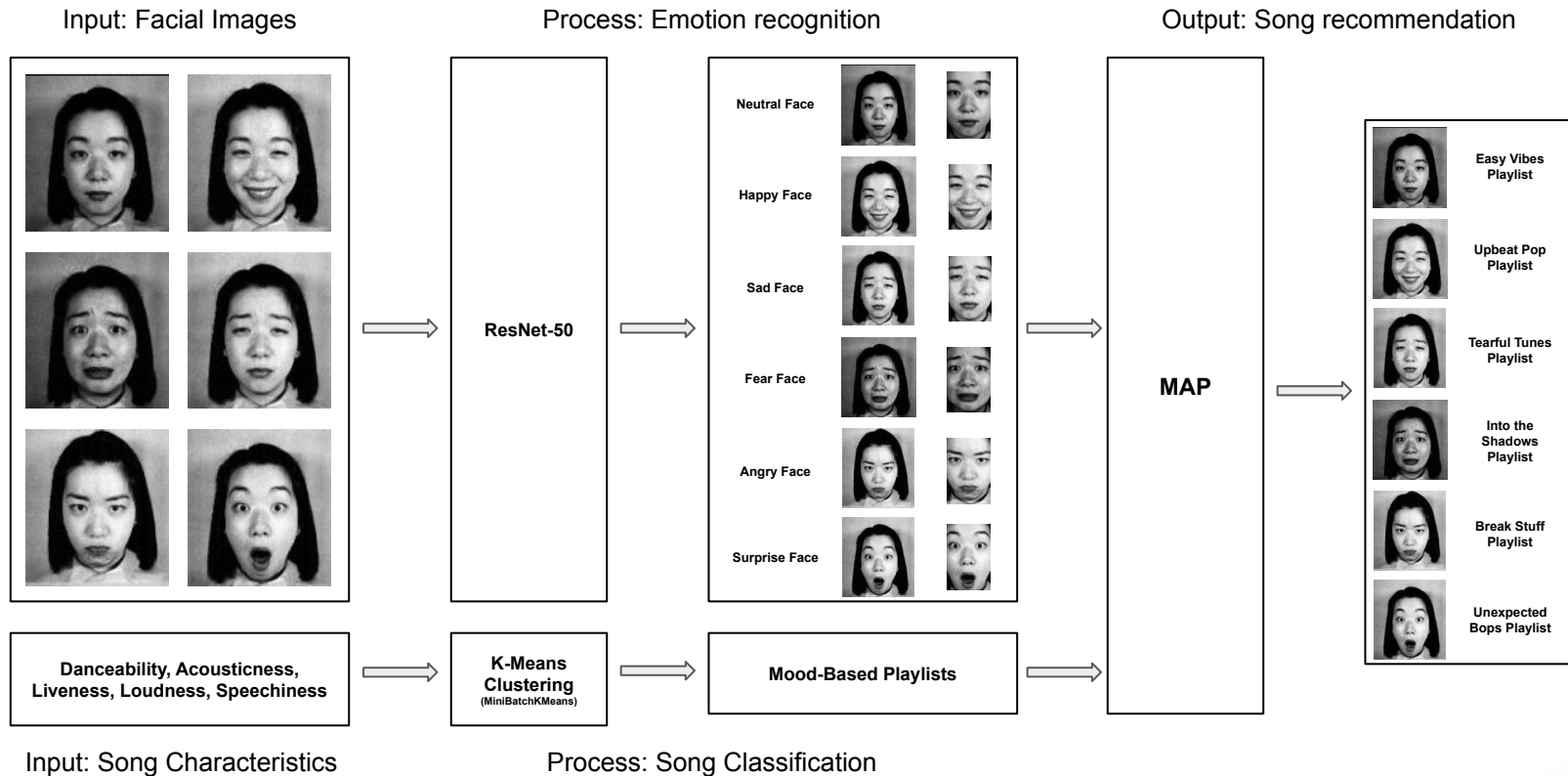
Identified Gap:

- Lack of dynamic, emotion-aware recommendation systems
- Need for integrating facial emotion analysis with personalization

Our Solution:

- Leverages facial emotion detection to recommend music tailored to the user's mood
- Employs MiniBatchKMeans clustering to classify songs by mood
- Enhances user engagement with adaptive and personalized recommendations

Project Overview and Objectives



Dataset and Preprocessing Techniques

FER-2013 Dataset

- 48x48 grayscale facial images
- 7 emotion categories: Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral
- 28,709 training images
- 3,589 test images

Preprocessing Techniques

- **Normalization:** Standardizes pixel values for consistency
- **Data augmentation:** Rotation, zoom, flipping, and more to reduce overfitting
- **Class balancing:** Computed class weights to handle imbalances in the dataset
- **Stratification:** Ensures each training batch has a balanced distribution of emotion categories to avoid bias in learning

Song Feature Extraction

- Extracted from songs to classify and recommend based on mood
- Features:
 - **Danceability:** Measures how suitable a song is for dancing
 - **Acousticness:** Likelihood of the song being acoustic
 - **Liveness:** Indicates how live the performance is
 - **Speechiness:** Measures the presence of spoken words in a song
 - **Loudness:** Overall loudness of the track



Model Architecture: ResNet-50 with Fine-Tuning

Why ResNet-50?

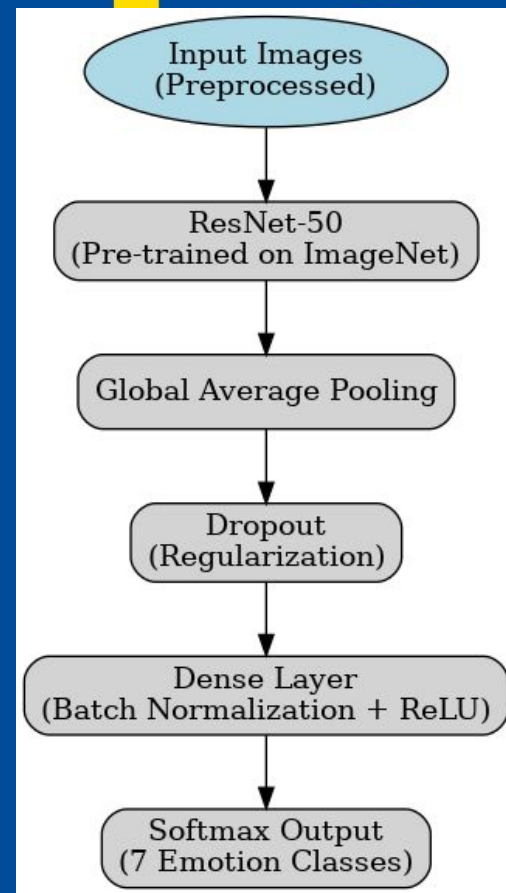
- Handles vanishing gradient problem with residual connections.
- Highly effective at capturing important patterns in complex images.
- Pre-trained on ImageNet for transfer learning.

Adaptations for Emotion Recognition:

- Leveraged pre-trained ResNet50 with frozen layers for feature extraction.
- Custom classification head added:
 - **Global Average Pooling** to reduce feature maps to dense representations.
 - Fully connected Dense layers with Batch Normalization.
 - Dropout layers to prevent overfitting.
 - Softmax output layer for 7 emotion classes.

Training Optimization Techniques:

- **Checkpoint:** Saves the best model during training based on the highest validation accuracy.
- **Early Stopping:** Stops training if validation accuracy doesn't improve after 7 epochs.
- **Learning Rate Schedule:** Reduces the learning rate by a factor of 0.5 if validation accuracy plateaus for 3 epochs.



Loss Function and Optimization for Image Classification

Loss Function:

Sparse Categorical Cross Entropy:

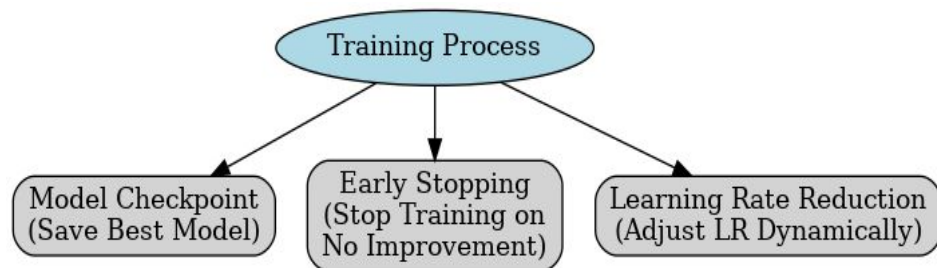
- Used for multi-class classification where labels are integers.
- Measures the difference between the true labels and predicted probabilities.

Optimizer:

Adam Optimizer

- Combines the benefits of momentum and adaptive learning rates for faster convergence.
- Learning rate set to **0.0005** for fine-tuned training.
- Integrated with callbacks for early stopping and learning rate reduction.

$$\text{Loss} = -\sum_{i=1}^N y_i \log(\hat{y}_i)$$



Model Architecture: K-Means Clustering for Song Classification

Why K-means Clustering?

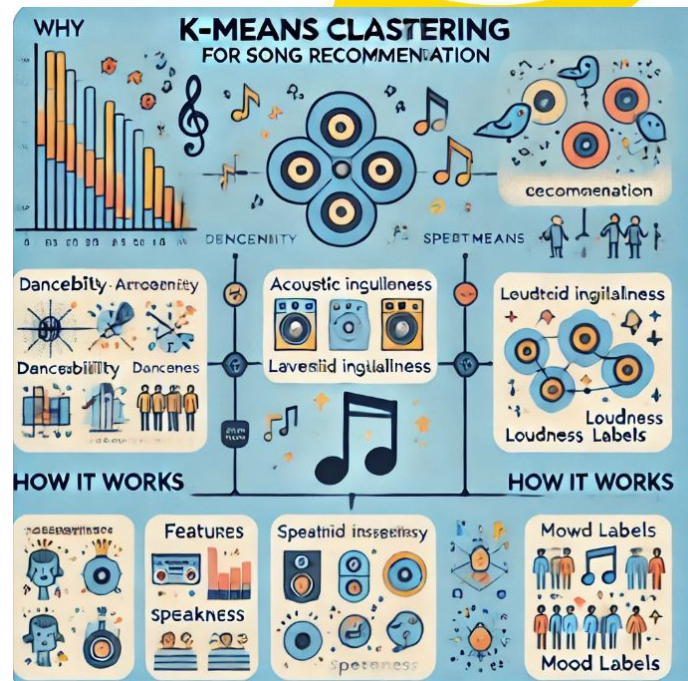
- Efficient and scalable for large datasets.
- Works well for grouping similar data points based on features.
- Offers interpretable outputs (feature importance) for song classification.

Features for Classification:

- Tempo (BPM): Song speed.
- Energy: Perceived intensity and activity.
- Danceability: Rhythm suitability for dancing.
- Valence: Positivity or negativity of the mood.
- Acousticness: Likelihood of being acoustic.
- Instrumentalness, Speechiness, Loudness, and Mode.

How It Works:

- Normalize the features to bring them into the same scale.
- Initialize KKK centroids randomly or using K-Means++.
- Assign each data point to the nearest centroid.
- Recalculate centroids as the mean of assigned data points.
- Iterate until convergence (when centroids stop changing).
- Map clusters to specific moods using manual or heuristic methods.
- Recommend songs from the same cluster based on detected mood.



Loss Function and Optimization for Song Recommendation

- Within-cluster sum of squares (WCSS) loss- The WCSS loss ensures that points are assigned to the closest cluster centroid.
- The aim is to minimize the variance of data points within each cluster.
- K-means++ initializes centroids to be distant from each other, improving convergence and better initial centroids.
- Ensures better clustering quality for highly diverse song features (e.g., acousticness, danceability).

Experimental Setup and Training

FER2013 Dataset:

- 35,887 images

Dataset split:

- Training (80%)
- Validation (20%)
- Stratification

Key hyperparameters:

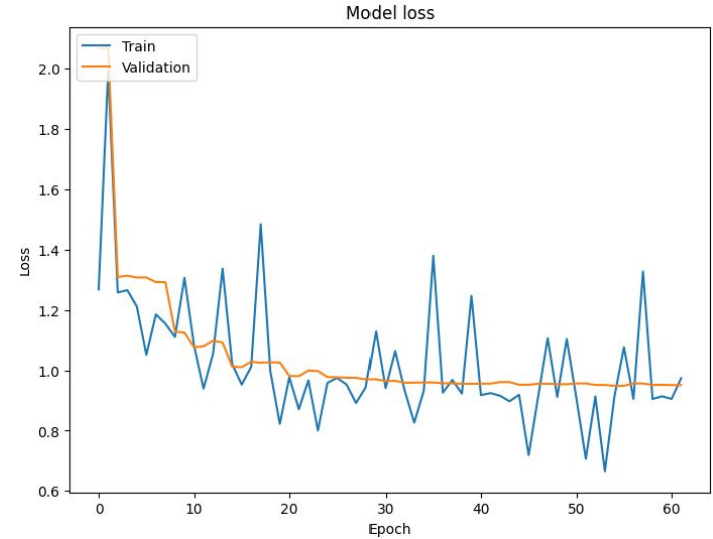
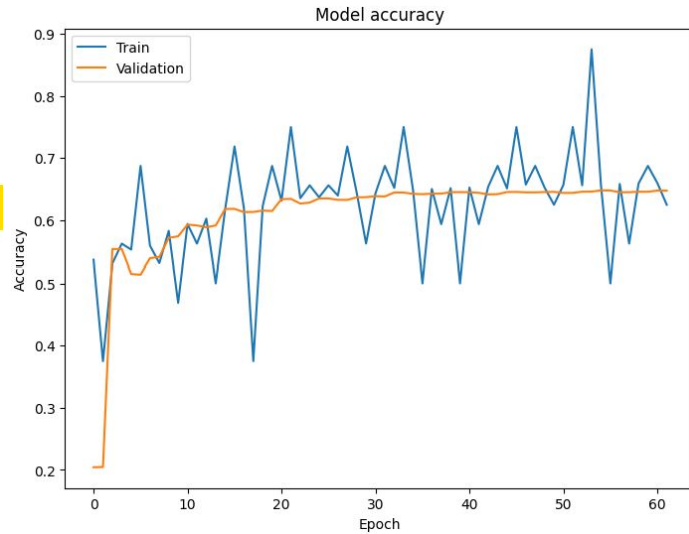
- Batch size: 32
- Epochs: 62
- Learning rate: 0.0005 with learning rate reduction

Training platform:

- GPU-accelerated environment

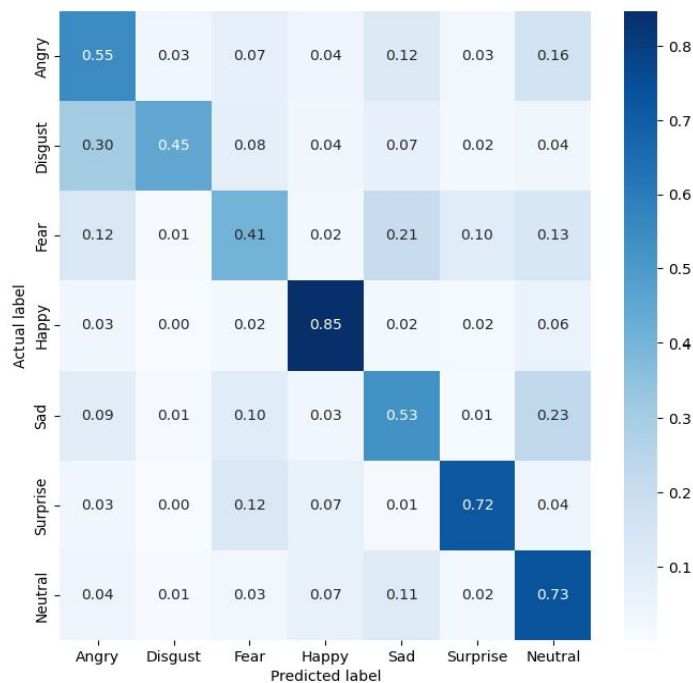
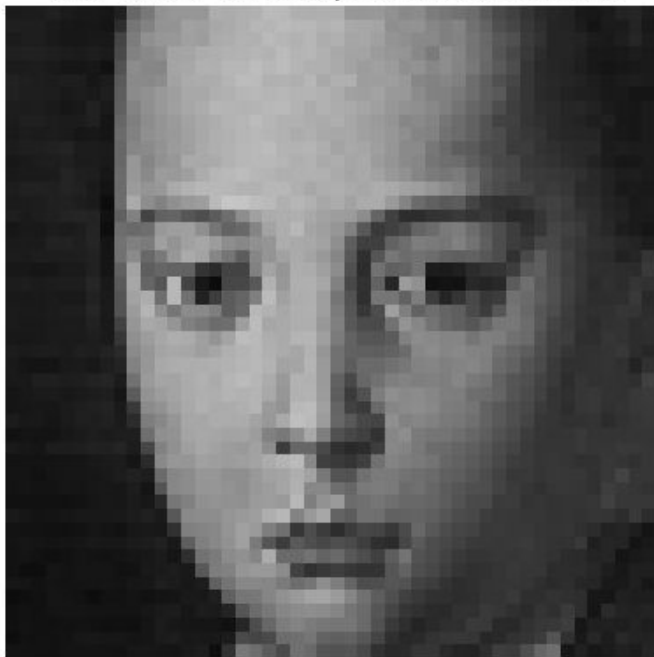
Model Performance and Metrics

Component	Train Accuracy	Validation Accuracy
ResNet-50 Emotion Detection	65.70%	64.81%



Model Performance and Metrics

True Label: Neutral | Predicted: Neutral



Figures
based on
ResNet-50

Predicted Probabilities: `[[4.9752422e-02 9.9862023e-05 5.3420417e-02 1.4858078e-03 3.3502978e-01
2.1571091e-03 5.5805463e-01]]`
Predicted Label: 6 (Neutral)

Challenges and Future Directions

Challenges faced:

- Dataset limitations (e.g., resolution, diversity)
- Trade-offs in fine-tuning depth

Future directions:

- Expanding dataset diversity
- Class balancing
- Incorporating user feedback loops for improved recommendations

Emotion	Count
Happy	8989
Neutral	6198
Sad	6077
Fear	5121
Angry	4953
Surprise	4002
Disgust	547

Conclusion

Project Summary:

- Integrated ResNet-50 for emotion detection and K-Means Clustering for song classification.
- System bridges real-time emotional context with personalized music recommendations.
- Successfully combined computer vision and machine learning for dynamic playlist generation.

Key Results:

- Emotion Detection: Achieved 64.81% accuracy with robust handling of minority classes.

Challenges Overcome:

- Addressed insufficient and imbalanced data in emotion detection with data augmentation.
- Preprocessing challenges addressed using image resizing and grayscale to RGB conversion.
- Tackled overlapping song categories using K-Means Clustering.
- Tackled overfitting by utilizing dropout, early stopping, and augmentation.
- Broader Implications:
 - Demonstrated the potential for AI to enhance user experience through emotion-aware systems.
 - Framework can be extended to other domains, such as healthcare or personalized content delivery.

Team Members' Contributions

Jessie Ma

- Image Classification Implementation
- Report and Presentation

Termeh Mohebbie

- Song Classification Implementation
- Report and Presentation

Rehan Ashfaq Syed

- Image Classification Implementation
- Research and Integration

Harsh Panchal

- Song Classification Implementation
- Research and Integration

Thank you!

